

COMPASS Synoptic FE 2024 Sample pH Issues

Background Information:

Field collection

- Synoptic FE samples are collected at each porewater collection from suction lysimeters for Fe(II) and Fe(III)
- Sample vials are trace metal acid washed, then in the lab 75 uL of 50% TCM HCl is added to the vial
 - o For reference, in industry typically 0.2mL of 50% HCl is used for 40mL of sample, which would mean for a 5mL sample you would only need 25 uL
- In the field, 5mL samples are filtered with a 0.45 um filter and collected into the prepped vials.
- The acidified samples should be around a pH of 2 after being collected into the prepped vial. Samples are kept in the fridge until analysis.
- The pH 2 should preserve the Fe(II)/Fe(III) in their current state and prevent oxidation or reduction prior to analysis

Laboratory Analysis

- We use a calorimetric method for analyzing iron
- It is critical for the method that the pH of the samples to be similar to the standards, which are at a pH of 2
- If sample pH is too high, colors will change too early, and it has been noted that a higher pH results in an inflated absorbance and therefore an inflated value

ISSUE

- Melanie noticed that a few samples she was running appeared to have a pH that was higher than 2 from the color changes
- She tested the pH, and they were closer to 5

Diagnosis:

